

Harsh Mohan Gajra

Senior Software Engineer

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Senior Software Engineer with 8+ years of solid experiences building for software development teams of industry cloud leaders like Amazon Web services (AWS) and dynamic AI data platform startups like Crux Data(backed by Goldman Sachs).

Highly experienced in all aspects of software development life-cycle and system design, from concept through development to delivery. Consistently recognized as a competent software engineer, with hard technical as well soft skills, required for coordinating cross-functional teams in a fast-paced environment to steer timely and quality project completion.

Equipped with a Master's in Computer Science from State University of New York at Binghamton, I am looking forward to obtaining the position of Technology Lead / Senior Software Engineer/ Staff engineer at a leading organization, to utilize my skills in software development to develop highly resilient and quality software products.

Education

Masters in Computer Science - State University Of New York, Binghamton, August 2016 - May 2018

Bachelor of Computer Engineering - K.J.Somaiya College of Engineering, Mumbai ,First class (2011- 2015)

Relevant Coursework: Data Structures and Algorithms, Database Systems, Operating Systems, Object Oriented Programming Languages, Design Patterns.

Technical Skills

Java 8 / 11	Amazon Web Services	GitHub
Spring Boot	Brazil	Cloud Development kit
Git	JIRA	RDS
Postgres	MySQL	DynamoDB
AWS Systems Manager	SWF	AWS Lambda
CI/CD	EC2	AWS Cloudwatch
Airflow	Google Cloud Platform	BigQuery
Docker	Python 3	Pub/Sub

Work Experience

Senior Software Engineer – Crux Data (Feb 2023 – Present)

Technology Stack: Python 3, FastAPI, SQLAlchemy, Java, Spring Boot, OpenAI (LLMs), GCP, Docker, Kubernetes, Airflow, PostgreSQL, Cassandra, BigQuery, GitHub Actions

- Developed an AI-powered suggestion engine leveraging OpenAI's GPT models to automatically recommend the optimal adapter and configurations based on file contents for ingestion pipelines executed via Apache Airflow.
- Prototyped and validated solutions utilizing large language models (LLMs) to intelligently process and normalize heterogeneous file formats, enhancing data standardization and downstream integration efficiency.
- Spearheaded a high-volume NoSQL-to-SQL migration project, transferring over 2 billion records from Cassandra to Google AlloyDB (PostgreSQL), employing Python, Bash, SQL, Kubernetes, BigQuery, and GCP—achieving an estimated \$300,000 in annual infrastructure cost savings.
- Built and maintained APIs with Python and FastAPI, using PostgreSQL and SQLAlchemy for data persistence, containerized via Docker, managed with Kubernetes, and deployed on GCP.
- Designed and deployed scalable APIs in Java with Spring Boot, utilizing PostgreSQL and Kubernetes on GCP.
- Engineered and orchestrated robust DAGs for data pipeline automation on both GCP-managed Cloud Composer and self-hosted Airflow clusters running on Kubernetes.
- Contributed to the setup and deployment of scalable Kubernetes pods to manage distributed applications.
- Architected a CI/CD-based Domain Synchronization tool that maps YAML workflow configurations directly into the backend domain model, triggered by GitHub Pull Requests. Integrated GitHub Actions to automate and streamline the synchronization pipeline, establishing Git as the authoritative source of truth for domain changes.

Software Engineer - Amazon Web Services, Data Migration Service (<https://aws.amazon.com/dms>)
Seattle, Washington (Sept, 2021 – Feb 2023)

Technology Stack : Java, AWS services like DMS, RDS, CDK, CloudFormation, Spring Boot

- Worked on designing and developing a Distributed Test Execution platform, that can manage and execute hundreds of thousands of test cases in parallel using serverless technologies like AWS Lambda, AWS Simple Workflow, AWS CloudWatch, Cloud development Kit and Java 11 to test different components AWS Data Migration Service like Control Plane, Data Plane(Kamino) and Serverless at scale.
- Worked on re-architecting the Canary and Test infrastructure to eliminate deployment bottlenecks and congestions leading to overall stabilization and faster deployments saving 50 hours a week.
- Started as 1st hire on the Platform STP team, mentored and lead a team of 3 interns and 2 engineers, and served as subject matter expert for multiple components of the software product.
- Wrote CI/CD pipelines using AWS Cloud development kit and CloudFormation to perform infrastructure deployments predictably and repeatedly, with rollback on error.
- Wrote automation, identified and eliminated bugs, fixed design flaws across the product stack concerning AWS DMS bringing down on call support tickets by 60 % and saving 2400 hours yearly.

Software Developer 2 – FINRA

Reston, Virginia (May 2020 - Sept, 2021)

Technology stack: Java, AWS Services

- Contributed to the design and development of FINRA's Consolidated Audit Trail project.
- Wrote SQL queries to pull data and develop reports on stocks and options that were used in Project Dashboards.
- Collaborated with external teams and stakeholders to determine requirements and drive product deliverables.
- Designed and implemented features for FINRA's Consolidated Audit Trail project which manages an extensive trail of transactions across different stock exchanges.

Software Engineer - Pype Inc (Acquired by AutoDesk in May 2020)

Herndon, VA, USA (July 2018 – May 2020)

Technology stack : Java 8, Spring Boot, MySQL, RabbitMQ

- Designed and developed internal software solutions for optimizing the business process followed by in-house Sales and Support teams thus saving the organization \$360,000 per year in costs and 15,000 man-hours per year.
- Designed the data model and developed the backend code for an Email Cadence software tool using Java 8, Spring Boot, JPA and MySQL database.
- Designed applications using the message brokers like RabbitMQ for message queuing and asynchronous job processing.
- Algorithm Development: Design, Analyze and implement algorithms in core Java for solving various use cases across the software products of the company.
- Wrote automations for the manual tasks performed by Customer Support and Success Teams saving our 1000 hours/year in the process.

Paper Publications

- **Paper:** Credit card Fraud Detection and Prevention Using Perceptron Training Algorithm.
- **Journal:** International Journal on Recent and Innovation Trends in Computing and Communication, Apr 2015
Link: <https://ijritcc.org/index.php/ijritcc/article/view/4182/4182>